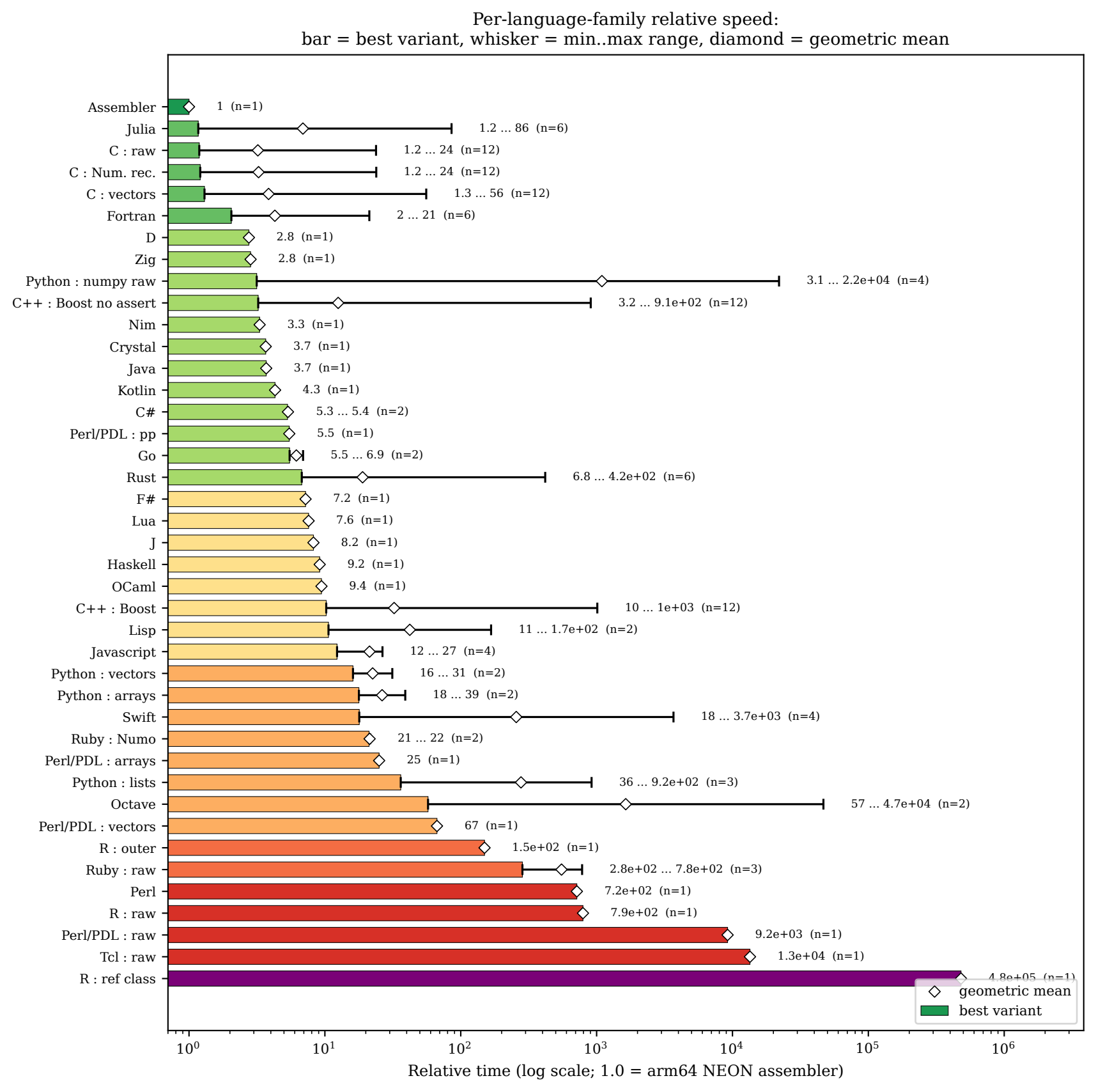
Per-language-family relative speed:
bar = best variant, whisker = min..max range, diamond = geometric mean
Assembler
Julia
C : raw
C : Num. rec.
C : vectors
Fortran
D
Zig
Python : numpy raw
C++ : Boost no assert
Nim
Crystal
Java
Kotlin
C#
Perl/PDL : pp
Go
Rust
F#
Lua
J
Haskell
OCaml
C++ : Boost
Lisp
Javascript
Python : vectors
Python : arrays
Swift
Ruby : Numo
Perl/PDL : arrays
Python : lists
Octave
Perl/PDL : vectors
R : outer
Ruby : raw
Perl
R : raw
Perl/PDL : raw
Tcl : raw
R : ref class
1 (n=1)
1.2 ... 86 (n=6)
1.2 ... 24 (n=12)
1.2 ... 24 (n=12)
1.3 ... 56 (n=12)
2 ... 21 (n=6)
2.8 (n=1)
2.8 (n=1)
3.1 ... 2.2e+04 (n=4)
3.2 ... 9.1e+02 (n=12)
3.3 (n=1)
3.7 (n=1)
3.7 (n=1)
4.3 (n=1)
5.3 ... 5.4 (n=2)
5.5 (n=1)
5.5 ... 6.9 (n=2)
6.8 ... 4.2e+02 (n=6)
7.2 (n=1)
7.6 (n=1)
8.2 (n=1)
9.2 (n=1)
9.4 (n=1)
10 ... 1e+03 (n=12)
11 ... 1.7e+02 (n=2)
12 ... 27 (n=4)
16 ... 31 (n=2)
18 ... 39 (n=2)
18 ... 3.7e+03 (n=4)
21 ... 22 (n=2)
25 (n=1)
36 ... 9.2e+02 (n=3)
57 ... 4.7e+04 (n=2)
67 (n=1)
1.5e+02 (n=1)
2.8e+02 ... 7.8e+02 (n=3)
7.2e+02 (n=1)
7.9e+02 (n=1)
9.2e+03 (n=1)
1.3e+04 (n=1)
4.8e+05 (n=1)
geometric mean
best variant
10^0
10^1
10^2
10^3
10^4
10^5
10^6
Relative time (log scale; 1.0 = arm64 NEON assembler)